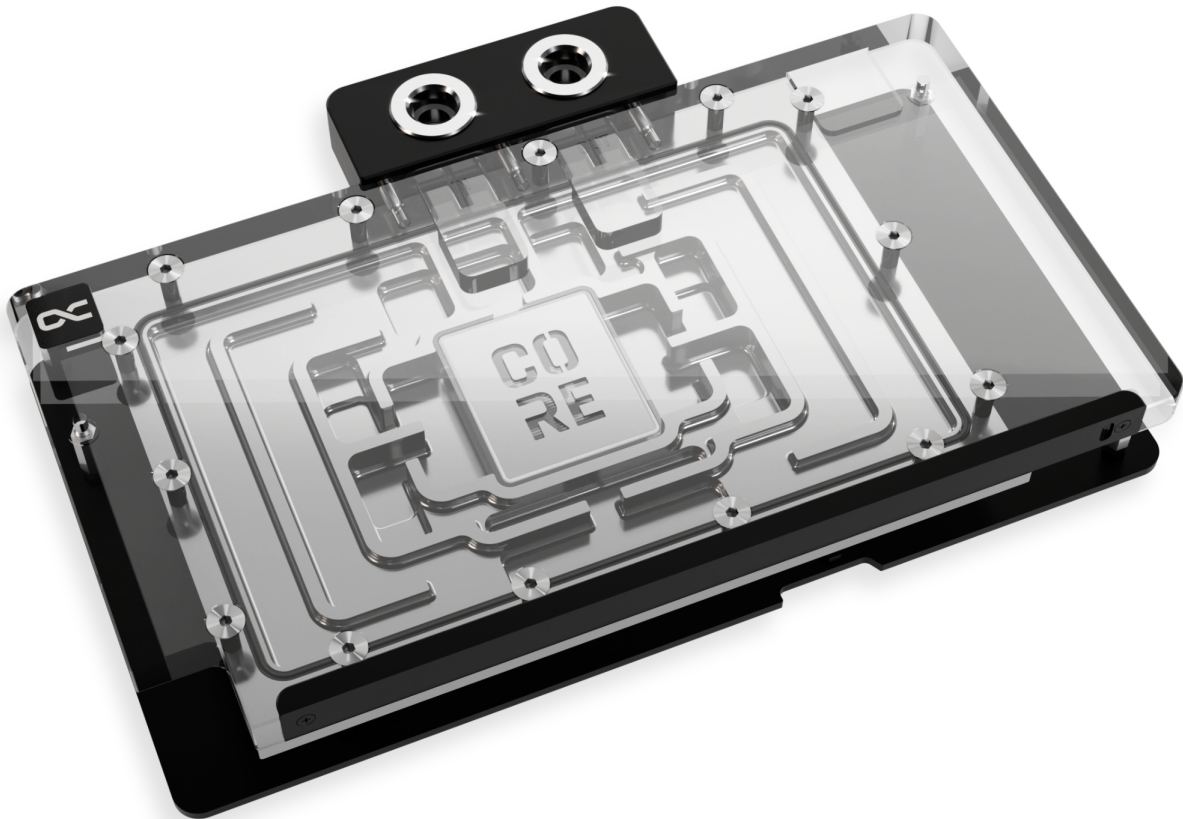


Alphacool Core Geforce RTX 5090 TUF Gaming mit Backplate

Alphacool article number: 10269



Quick Info

Alphacool introduces its new series of graphics card water coolers for the latest Nvidia GeForce RTX generation. Part of the Core product series, these coolers combine the highest quality with exceptional performance. Thanks to an optimized technical design, these GPU coolers deliver outstanding cooling performance.

- Optimized water flow through software and practical measurements
- Massive brass threads for secure fastening
- Superior chrome plating instead of nickel plating on the copper cooler
- Core design for maximum quality and feel

Compatibility

- Asus TUF Gaming GeForce RTX™ 5090 32GB GDDR7 (90YV0LY1-M0NA00)
- Asus TUF Gaming GeForce RTX™ 5090 32GB GDDR7 OC Edition (90YV0LY0-M0NA00)

Scope of delivery

1x Core Geforce RTX 5090 TUF Gaming cooler	1x Thermal Grease
1x Backplate	1x Putty tool
2x 68x15x1mm Pad	1x ARGB Adapter
1x 54x15x1mm Pad	1x PCIe Bracket
4x 8x8x1mm Pad	1x Plug tool
2x 13x15x1mm Pad	1x Screwdriver
1x 103x8x1mm Pad	2x G1/4" Plug
1x 112x8x1mm Pad	6x M2x10mm screws
2x 68x8x3mm Pad	1x M2x5mm spring screw
1x 45x45x3mm Pad	1x washer
1x 20x8x3mm Pad	1x M2x5mm screw
1x 54x8x3mm Pad	

Technical data cooler

Dimensions total (L x W x H)	235,95 x 155,57 x 34,12mm
Material cooler	chrome-plated copper
Material cooler top	Acrylic
Material terminal	Nylon
Material Connection thread	Brass
Max working temperature	60°C
Pressure tested	0,8 bar
Threads	4 x G1/4"
Thickness cooler bottom	3,25mm
Thickness cooling fins	0,4mm
Distance cooling fins	0,4mm
Illumination	digital aRGB LEDs
Power connector digital aRGB LEDs	3-Pin JST
Power digital aRGB LEDs	5V
Number of digital aRGB LEDs	9
Color	transparent

Technical data backplate

Dimensions (L x W x H)	235,95 x 127,00 x 6mm
Thick backplate without spacers	3mm
Material	Aluminium

Download links

Manual	10269_Alphacool_Core_Geforce_RTX_5090_TUF_Gaming_mit_Backplate_Manual.pdf
Product pictures	10269_Alphacool_Core_Geforce_RTX_5090_TUF_Gaming_mit_Backplate_pics.zip
EU-DoC / CE / RoHS	Alphacool_EuDoc_CE_RoHS_10269.pdf

Packaging dimensions per unit

L x W x H	350 x 205 x 60 mm
Weight	1700 g

Other data

Certificates	CE, FC, RoHS
EAN	4250197102698
Customs code	84195080900
Guarantee	3 years

Article text

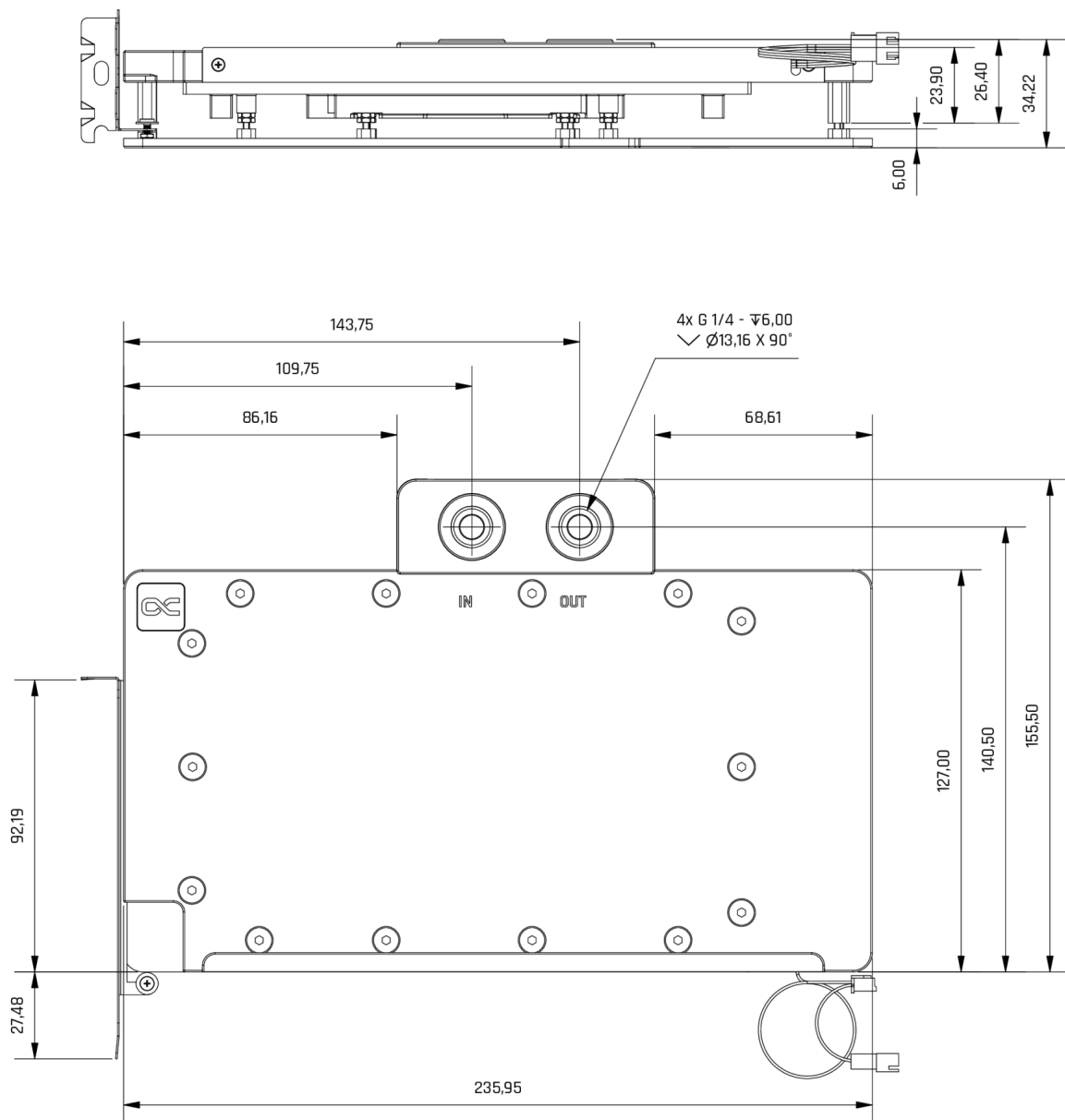
Alphacool introduces its new series of graphics card water coolers for the latest Nvidia GeForce RTX generation. Part of the Core product series, these coolers combine the highest quality with exceptional performance. Thanks to an optimized technical design, these GPU coolers deliver outstanding cooling performance.

At the heart of the cooler is a copper block, precisely milled from a single piece and complemented by a precisely crafted brass terminal with a nylon cover. A chrome finish enhances the copper cooler's durability and provides a smoother, more radiant surface than traditional nickel plating.

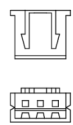
The visual design of the GPU water cooler has been slightly modified. A new highlight is the Core emblem placed above the jetplate, which emphasizes its affiliation with the Core series and visually enhances the design. The series continues to feature high-quality brass connectors that ensure optimal quality and safety. The specially adapted aluminum backplate covers the entire backside of the copper cooler, ensuring a clear and uniform appearance. The acrylic, combined with aRGB lighting, allows for even and impressive illumination of the entire cooler.

The GPU water cooler has been completely overhauled technically. To optimize cooling performance, the distances between the cooler and the components of the PCB were redefined and tested. The water flow was simulated using the most advanced software and verified through practical tests. These extensive tests allowed for improvements to the cooler base and jetplate, ensuring the best possible cooling performance for the new Nvidia GeForce RTX generation.

Figure (1)



aRGB
Connection



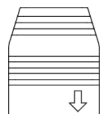
VCC 5V DC
DATA
GND

aRGB
Socket



GND
DATA
VCC 5V DC

aRGB
Connection



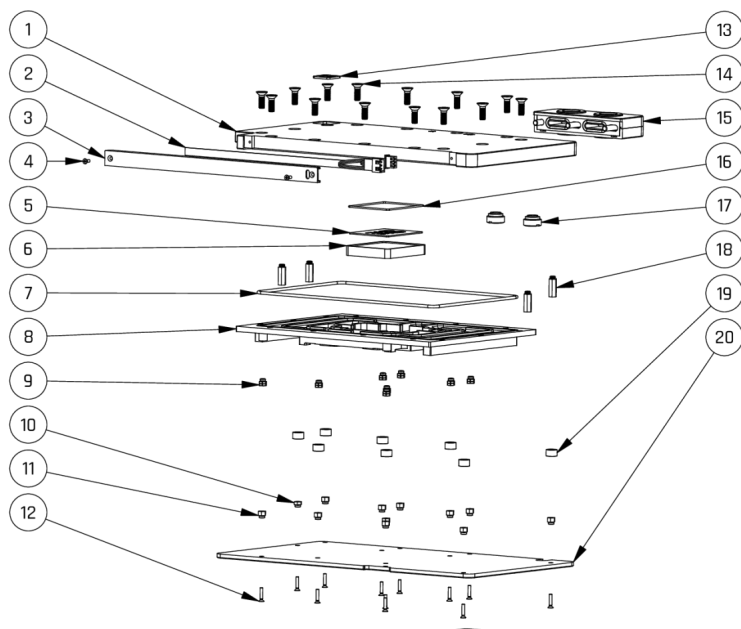
GND
DATA
VCC 5V DC

General tolerance: $\pm 0,25\text{mm}$
Thickness with graphics card (PCB)
Dimension in millimeter

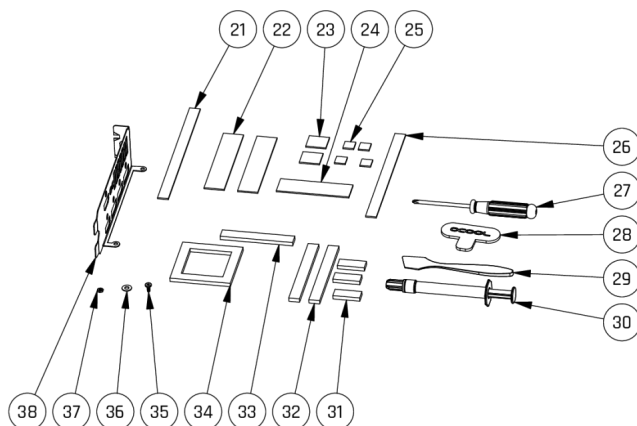
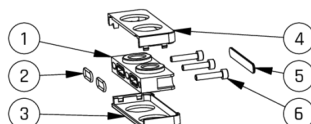
Figure (2)

Mechanical properties:

LED quantity and length	10pc. / L =166mm	Working pressure p =	0,0 - 1,0Bar	Maximum torque of M max =	
LED Voltage V =	5 V	Maximum pressure p max =	2,0Bar	M thread	M2X0,4 = 0,13Nm
LED Ampere A =	0,06 A	Maximum torque of G thread M max =	G1/4 = 1,0Nm	DIN 13-1 Metric ISO standard thread	M4X0,7 = 1,12Nm
LED Power W =	3 W	ISO 228-1 Parallel Whitworth pipe thread BSPP (GB)			
Minimum temperature t min =	-20°C				
Working temperature t =	20°C				
Maximum temperature t max =	60°C				



No: 15



Cooler - Spare parts list

Number	Quantity	Description	Material
1	1	Top	LXN865
2	1	LED strip / 10 LEDs	-
3	1	LED Bracket	AlMg0,7Si
4	2	ISO 7046-1 - M2 x 5	Cr01
5	1	Core-Jet-Logo	CuZn39Pb3
6	1	Jet	PMMA Acryrex®
7	1	ORing 164 x 2	NBR Shore 70
8	1	Copper	Cu-HCP
9	8	GPX Spacer SW5 M4X0,7-M2X0,4 x 3	CuZn39Pb3
10	1	CORE Spacer SW5 M4x0,7 x 2	PA 66,GF25 FR(V-2)
11	11	CORE Spacer SW5 - M4X0,7 x 3	PA 66,GF25 FR(V-2)
12	12	ISO 7046-1 - M2 x 10	X10CrNi18-8
13	2	Alphacool Logo Batch 15x15mm 2023	
14	14	ISO 10642 M4 x 12	X10CrNi18-8
15	1	AC CORE Terminal	
16	1	Core-Jet-Logo O-Ring 48,5 x 1	VMQ Shore A 70
17	2	Sealing plug V.2 G1_4 Chrome	CuZn39Pb3
18	4	GPX Spacer SW5 M4X0,7-M2X0,4 x 13	X10CrNi18-8
19	8	PCB Spacer	PA 66,GF25 FR(V-2)
20	1	Backplate	AlMg1SiCu

Terminal - Spare parts list

Number	Quantity	Description	Material
1	1	Terminal	CuZn39Pb3
2	2	ORing 13,5 x 1,5	NBR Shore 70
3	1	Bottom Cover	PA 66,GF25 FR(V-2)
4	1	Cover	PA 66,GF25 FR(V-2)
5	1	RTX Label	PA MACM12 + PA 12, GHLT, 18-020
6	3	ISO 4762 - M4 x 20	X5CrNi18-10

Cooler accessories - Spare parts list

Number	Quantity	Description	Material
21	1	Pad 112 x 8 x 1	7W/mK - Shore-00 35
22	2	Pad 68 x 15 x 1	7W/mK - Shore-00 35
23	2	Pad 13 x 15 x 1	7W/mK - Shore-00 35
24	1	Pad 54 x 15 x 1	7W/mK - Shore-00 35
25	4	Pad 8 x 8 x 1	7W/mK - Shore-00 35
26	1	Pad 103 x 8 x 1	7W/mK - Shore-00 35
27	1	Screwdriver	-
28	1	Plug Tool	PA 66,GF25 FR(V-2)
29	1	Putty Tool	-
30	1	Alphacool Subzero Thermal grease 1g	-
31	3	Pad 20 x 8 x 3	7W/mK - Shore-00 35
32	2	Pad 68 x 8 x 3	7W/mK - Shore-00 35
33	1	Pad 54 x 8 x 3	7W/mK - Shore-00 35
34	1	PAD 45 x 45 x 3	7W/mK - Shore-00 35
35	1	GPX Flat head screw M2X0,4 x 5	X10CrNi18-8
36	1	ISO 7089 - 3 PA66	PA 66,GF25 FR(V-2)
37	1	ISO 4035 M2	X10CrNi18-8
38	1	Slot Bracket	Cr01